

PROJECT DEVELOPMENT PHASE



Coding & Solution

 **DATE**
17 7/9/2026

 **NAME**
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 **PROJECT**
**OptiCrop – Smart
Agricultural
Production
Optimization Engine**

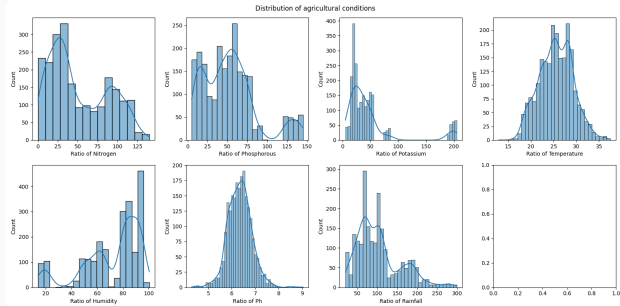
 **MAX MARKS**
5 Marks

Overview

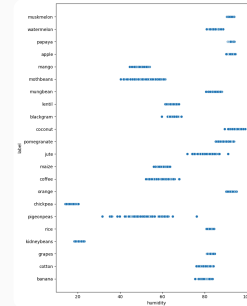
The OptiCrop machine learning pipeline and web application were implemented in Python. The pipeline below reflects the end-to-end coding approach, from raw data to a running Flask app.

Development Pipeline

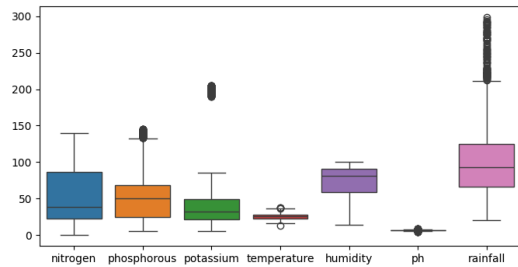
STAGE	FILE	WHAT IT DOES
Dataset generation	data/generate_dataset.py	Builds Crop_recommendation.csv (2200 rows, 22 crops) with realistic soil and climate ranges per crop.
EDA & preprocessing	model/train_model.py	Loads the dataset, runs univariate/bivariate/multivariate analysis, checks nulls, and removes outliers using the IQR method.
Pattern discovery	model/train_model.py	Applies K-Means clustering with the elbow method to group similar soil/climate conditions.
Model training	model/train_model.py	Trains a Logistic Regression classifier on the 7 soil/climate features and evaluates it with a classification report.
Model persistence	model/model.pkl	The trained model is serialized with pickle so the web app can load it without retraining.
Web backend	app/app.py	Flask routes for Home, About, and Find your crop; loads model.pkl and returns predictions from form input.
Frontend	app/templates/*.html, app/static/style.css	Responsive HTML/CSS pages for navigation and the soil/climate input form.



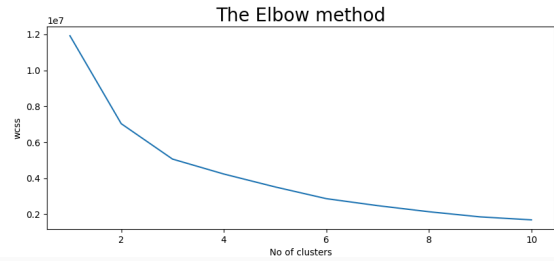
Univariate distribution of soil & climate features



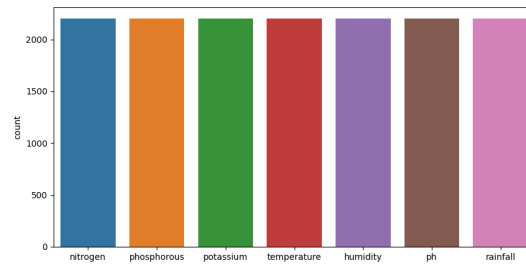
Bivariate analysis: humidity vs crop label



Boxplot before outlier removal



Elbow method for K-Means clustering



Feature countplot across crop classes

Model Performance

98%
TEST ACCURACY

400
HELD-OUT SAMPLES

20
CROP CLASSES